# A Generative Model for a Dataset of Greyscale Images

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April 26, 2026

**Abstract**

Neural scaling laws describe the relationship between the performance of large neural networks and dataset size, acting as an effective guideline for optimal training of LLMs among other models. This paper proposes a generative model for a dataset of greyscale images that adheres to a power-law scaling in the spectra of the covariance matrix. We show that the model is easily analyzable and derive a closed-form solution for the eigenvalues of the covariance matrix that is dependent on a single-variable function.

## 1 Introduction

Empirically, the population loss of neural networks has been observed to follow a power law relationship with regards to the number of parameters and training dataset size for various models and datasets. These relationships have therefore been referred to as *neural scaling laws* and specifically refer to the power law relation between loss and dataset size or parameter number regardless of model or setting. In the wake of LLMs and other large models, the general learning problem used to train neural networks is no longer necessarily constrained by the quantity of data but rather by the compute budget; a function of the dataset size, among other parameters [5, 1]. It follows that studying the relationship between the dataset size and loss allows for large neural networks to be more optimally trained. This paper proposes a deterministic generative model for a greyscale image dataset

based on the MNIST dataset that exhibits power-law scaling in the spectra of the covariance matrix. Additionally, the model and resulting dataset are easily analyzed, implemented and scalable for datasets with different image sizes from MNIST with closed-form solution for the resulting spectra of the covariance matrix.

# 2 Preliminaries

To start, consider a dataset X with T samples, where $x_\tau$ is some sample for $\tau \in \{1, ..., T\}$ that has been pre-emptively centred in a preprocessing stage. Suppose $x_{i;\tau}$ denotes the components of a given sample with $i = 1, ..., N$. Each $i$ indexes a different input feature of a sample which for example, might represent a pixel in an image. According to [4], the eigenvalues of the empirical covariance matrix given by

$$\frac{1}{T}\sum_{\tau=1}^{T} x_{i;\tau}x_{j;\tau} \tag{1}$$

are well fit by a power law distribution

$$\lambda_i \sim \frac{1}{i^{1+\alpha}} \tag{2}$$

for $i > 10$ for natural datasets. Moreover, this power law is known as a neural scaling law due to the fact that $\lambda_i \to 0$ as $i \to \min\{N, T\}$. As a result, it can be seen that as the number of samples in the dataset increases, the number of nonzero eigenvalues increases up to $N$.

Consequently, we find that the eigenvalues of natural datasets, and more specifically, MNIST, can be modeled using an artificial dataset created by a deterministic method that loosely distributes eigenvalues according to the power law. Furthermore, the covariance matrix of the artificial dataset is easily analyzable and contains circulant structure related to its deterministic generation method that allows for a closed-form solution to the eigenvalues of the covariance matrix.

# 3 Methods

We approximate the MNIST dataset using an artificial dataset with a fixed number of unique greyscale images. Each image in the artificial dataset

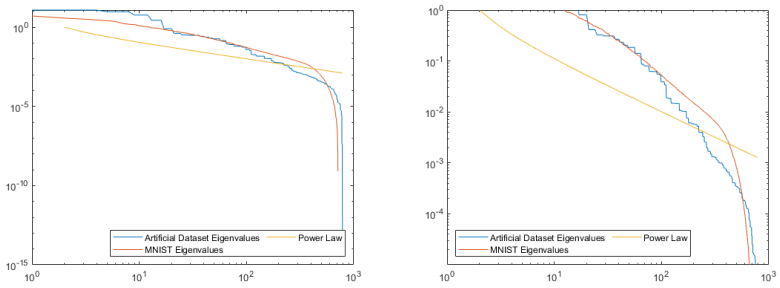



Figure 1: Eigenvalues of the MNIST Dataset, Artificial Dataset and the power law.

features a white horizontal and vertical line of 3-pixel width with black background and a single pixel width of greyscale on either side of the lines. As a result, the dataset features 784 28x28 images where each line is centered at a given row index and column index and each combination of row and column indices is unique. Additionally, in the case where the width of a line overlaps with the boundary of the image, the line is wrapped to the other side of the image as shown in Figure 2. Overlapping pixels between horizontal and vertical lines are assigned values equal to the max value of the two overlapping lines.

By deterministically using the code in Listing 1 to produce a single sample per combination of row and column indices, the dataset of 784 samples formed by each combination of row and column indices is fully deterministic and therefore allows for easier analysis while also being representative of a larger dataset where each class would have an equal number of samples per class. Since the covariance matrix of a dataset formed by producing a single sample per combination of indices is equal to a dataset formed by producing $x$ samples per combination of indices, the properties of the dataset generated are independent of the size of the dataset. Let $\widetilde{X}$ be the artificial dataset produced using the procedure described.

Informally, $\widetilde{X}$ somewhat approximates the MNIST dataset given that the greyscale, and line widths in samples of $\widetilde{X}$ generally match those of the drawn digits in the MNIST dataset despite the fact that one dataset features solely horizontal and vertical lines, and the other, drawn digits. More formally, $\widetilde{X}$ exhibits a covariance matrix with eigenvalues loosely following the power law given in [4] with more analyzable form as seen in Figure 1. Specifically, the

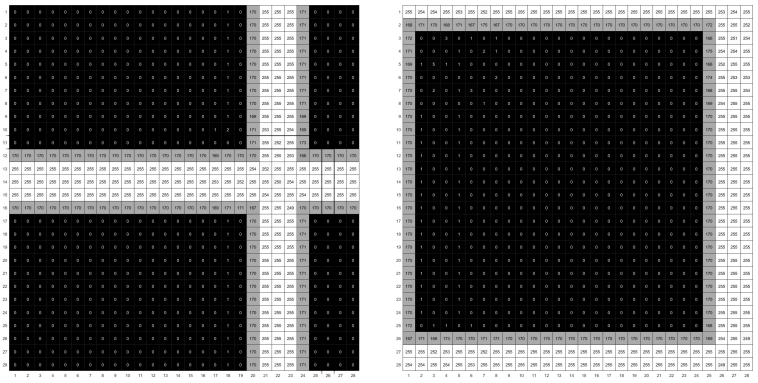


Figure 2: Samples from the artificial dataset. Left: row index=13, column index=21. Right: row index=27, column index=26.

covariance matrix of $\widetilde{X}$ is block circulant circulant block. Since images are typically flattened in column-major order, it can be seen that the columns of samples of $\widetilde{X}$ induce block circulant covariance and, in conjunction with the rows of samples of $\widetilde{X}$, result in block circulant circulant blocks.

# 4 Main Results

## 4.1 Proof of Block Circulant Circulant Blocks

Consider the distribution of values for samples in $\widetilde{X}$. We refer to the covariance matrix of the dataset produced by taking solely the j-th columns of each sample in $\widetilde{X}$ as the shortened covariance matrix for some $j$. Clearly, the mean of any sample in $\widetilde{X}$ is equal to the mean of any other sample due to the fact that regardless of the indices of the center of the horizontal and vertical lines, the values of the pixels of the sample will sum to the same value. This results directly from the procedure in which $\widetilde{X}$ was created and the fact that the line widths and greyscale wrap to the other side of the samples when intersecting with the edge. Therefore, the pre-centering has no effect on the analysis of the covariance matrix. By the same reasoning, the distribution of values for all of the k-th entries of the p-th columns in $\widetilde{X}$ is equal to the q-th entries of the j-th columns in $\widetilde{X}$ for $j, k, q, p = 1, ..., 28$. Since the i-th

diagonal entry of the covariance matrix is essentially calculated by taking the squared norm of the i-th entries of $\widetilde{X}$, it follows that all the entries of the diagonal of the covariance matrix are equal to one another. Similarly, if $\widetilde{X}$ is sliced to only include the j-th column of each sample, the resulting shortened covariance matrix also exhibits the same behavior. Since the distribution of values for a given entry of the j-th column is equal to the distribution of values for any other, the diagonal of the shortened covariance matrix must also be constant.

Next, consider the shortened covariance matrix $A$ for some column j and the entries $a_{i,i+1}$, and $a_{i+1,i+2}$. Due to the line widths and greyscale, in most cases, the i-th and i+1-th entry correlate positively. To be precise, the only case where they do not correlate positively is when the i-th entry has greyscale pixel value and the i+1-th has black pixel value. In these cases, there are equally many cases where the i-th entry has black pixel value and the i+1-th has greyscale pixel value irrespective of the value of i. Consequently, given that the distribution of values of all the i-th entries of the j-th column of every sample in $\widetilde{X}$ is equal to that of the distribution of values of any other distinct entry, all the entries of the form $a_{i,i+1}$ and $a_{i+1,i}$ must also have constant value. Naturally, this argument applies for entries of the form $a_{i,i+2}$ and $a_{i+2,i}$, $a_{i,i+3}$ and $a_{i+3,i}$, and $a_{i,i+4}$ and $a_{i+4,i}$ where the larger the gap in indices, the smaller the entry in the covariance matrix. However, after a gap of 4 in the indices, the values remain constant due to the fact that the line width of the horizontal and vertical lines is at most 5, including greyscale values. Therefore, all the shortened covariance matrices are circulant and all the block matrices in the covariance matrix of $\widetilde{X}$ are circulant. Following a similar argument, we see that the presence of horizontal lines causes the presence of block circulant behavior in the covariance matrix as a consequence of adjacent horizontal pixels having index separated by $n = 28$.

## 4.2 Derivation and Results

We start by noting the structure of the empirical covariance matrix of the artificial dataset. In particular, the empirical covariance matrix, A, is block circulant with circulant blocks ($BCCB$) such that

$$A = \begin{bmatrix} A^{(1)} & A^{(2)} & \cdots & A^{(n)} \\ A^{(n)} & A^{(1)} & \cdots & A^{(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ A^{(2)} & A^{(3)} & \cdots & A^{(1)} \end{bmatrix}$$

where $A^{(p)}$ is the p-th circulant block matrix of A and

$$A^{(p)} = \begin{bmatrix} a_0^{(p)} & a_1^{(p)} & \cdots & a_{n-1}^{(p)} \\ a_{n-1}^{(p)} & a_0^{(p)} & \cdots & a_{n-2}^{(p)} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{(p)} & a_2^{(p)} & \cdots & a_0^{(p)} \end{bmatrix}$$

for $p \in \{1, 2, ..., n\}$. Furthermore, each block matrix is also symmetric such that $a_q^{(p)} = a_{n-q}^{(p)}$ for $q \in \{0, 1, ..., n-q\}$. As a consequence of the dataset generation method**, block matrices closer to the block diagonal of A exhibit larger positive values that those farther away. Similarly, the block matrix entries closer to the diagonal of the block matrix also exhibit larger positive values. Lastly, we find that $A^{(6)} = A^{(7)} = ... = A^{(n-4)}$ and $a_6^{(p)} = a_7^{(p)} = ... = a_{n-5}^{(p)}$ as a result of the line width used for the dataset generation. Since A is *BCCB*, by Theorem 5.8.1, [3], A is diagonalizable by the discrete 2d fourier transform and therefore has eigenvectors equal to the columns of the 2d DFT matrix. Similarly, by Theorem 3.3.2, [3], circulant matrices are diagonalized by the DFT matrix and thus, a closed-form solution exists for the eigenvalues of circulant matrices.

For the sake of consistency, assume all matrix entries are indexed by $\mathbb{Z}_n$. Suppose $\Lambda$ is the diagonal matrix of eigenvalues of $A$ and $\Lambda_p$ is the diagonal matrix of eigenvalues of $A^{(p)}$ for $p = 1, ..., 28$. $\Lambda$ is given by

$$\Lambda = \sum_{k=0}^{n-1} \Omega^k \otimes \Lambda_{k+1} \tag{3}$$

where $\Omega = diag(1, \omega, ..., \omega^{n-1})$, $\omega = e^{\frac{2\pi i}{n}}$ and $\otimes$ is the kronecker product. For some arbitrary circulant matrix, C, the j-th eigenvalue is

$$\lambda_j = c_0 + c_1\omega^j + ... + c_{n-1}\omega^{(n-1)j} \quad \text{for } j = 0, ...., n-1 \tag{4}$$

where

$$C = \begin{bmatrix} c_0 & c_1 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & \cdots & c_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & \cdots & c_0 \end{bmatrix}$$

Let $\lambda_j^{(p+1)}$ be the $j-th$ eigenvalue of $A_{p+1}$ and $i = kn + j$ for some $k$, $j = 0, ..., n-1$. Combining the results from the two theorems, we obtain the following closed-form expressions for the $i-th$ eigenvalue $\lambda_{(k,j)}$

$$\lambda_{(k,j)} = \sum_{p=0}^{n-1} \sum_{q=0}^{n-1} \omega^{kp+qj} a_q^{(p+1)} \tag{5}$$

Given the equality between the block matrices with index 6 through $n-4$, and the equality between the entries of the block matrices with index 6 through $n-4$, rearranging unique pairs of values allows for the summations to be rewritten in more analyzable form. Moreover, taking advantage of the fact that $\omega^x + \omega^{-x} = 2\cos(\frac{2\pi x}{n})$ reveals that the inner summation is symmetric about $\frac{n}{2} - 1$. Lastly, we define $\sigma(k)$ to be 1 if $k$ even and $-1$ if $k$ odd for $k \in \mathbb{N}$. With these facts in mind, the expression can be expanded somewhat non-trivially.

$$\lambda_{(k,j)} = a_0^{(1)} + 2\cos\left(\frac{2\pi k}{n}\right) a_1^{(1)} + 2\cos\left(\frac{4\pi k}{n}\right) a_2^{(1)} + 2\cos\left(\frac{6\pi k}{n}\right) a_3^{(1)} + 2\cos\left(\frac{8\pi k}{n}\right) a_4^{1)}$$
$$+ \sigma(j) a_{\frac{n}{2}}^{(1)} + \sum_{q=5}^{\frac{n}{2}-1} 2\cos\left(\frac{2\pi jq}{n}\right) a_q^{(1)} + \sigma(k)\left(a_0^{(\frac{n}{2}+1)} + 2\cos\left(\frac{2\pi k}{n}\right) a_1^{(\frac{n}{2}+1)}\right.$$
$$+ 2\cos\left(\frac{4\pi k}{n}\right) a_2^{(\frac{n}{2}+1)} + 2\cos\left(\frac{6\pi k}{n}\right) a_3^{(\frac{n}{2}+1)} + 2\cos\left(\frac{8\pi k}{n}\right) a_4^{\frac{n}{2}+1)}$$
$$\left. + \sigma(j) a_{\frac{n}{2}}^{(\frac{n}{2}+1)} + \sum_{q=5}^{\frac{n}{2}-1} 2\cos\left(\frac{2\pi qj}{n}\right) a_q^{(\frac{n}{2}+1)}\right) + \sum_{p=1}^{4}\left(2\cos\left(\frac{2\pi kp}{n}\right)\left(a_0^{(p+1)}\right.\right.$$
$$+ 2\cos\left(\frac{2\pi k}{n}\right) a_1^{(p+1)} + 2\cos\left(\frac{4\pi k}{n}\right) a_2^{(p+1)} + 2\cos\left(\frac{6\pi k}{n}\right) a_3^{(p+1)} + 2\cos\left(\frac{8\pi k}{n}\right) a_4^{(p+1)}$$
$$\left.\left. + \sigma(j) a_{\frac{n}{2}}^{(p+1)} + \sum_{q=5}^{\frac{n}{2}+1} 2\cos\left(\frac{2\pi jq}{n}\right) a_q^{(p+1)}\right)\right) + \left(a_0^{(6)} + 2\cos\left(\frac{2\pi k}{n}\right) a_1^{(6)}\right.$$
$$+ 2\cos\left(\frac{4\pi k}{n}\right) a_2^{(6)} + 2\cos\left(\frac{6\pi k}{n}\right) a_3^{(6)} + 2\cos\left(\frac{8\pi k}{n}\right) a_4^{(6)} + \sigma(j) a_{\frac{n}{2}}^{(6)}$$
$$\left. + \sum_{q=5}^{\frac{n}{2}+1} 2\cos\left(\frac{2\pi j(q)}{n}\right) a_{q+5}^{(6)}\right)\left(\sum_{p=5}^{\frac{n}{2}+1} 2\cos\left(\frac{2\pi kp}{n}\right)\right) \tag{6}$$

Using the Dirichlet Kernel [2], we restructure nearly all the sums to obtain the final derivation.

$$\phi(k) \coloneqq \begin{cases} \left(\frac{\sin\left(\frac{(n-1)\pi k}{n}\right) - \sin\left(\frac{9\pi k}{n}\right)}{\sin\left(\frac{\pi k}{n}\right)}\right) & \text{for } k = 1,\ ..., n-1 \\ n-6 & \text{for } k = 0 \end{cases} \tag{7}$$

$$\lambda_{(k,j)} = \sum_{q=0}^{4} 2\cos\left(\frac{2\pi jq}{n}\right) a_q^{(1)} + \sigma(j)a_{\frac{n}{2}}^{(1)} + \phi(j)a_5^{(1)} + \sigma(k)\left(\sum_{q=0}^{4} 2\cos\left(\frac{2\pi jq}{n}\right) a_q^{(\frac{n}{2}+1)}\right.$$
$$\left. + \sigma(j)a_{\frac{n}{2}}^{(\frac{n}{2}+1)} + \phi(j)a_5^{(\frac{n}{2}+1)}\right) + \sum_{p=1}^{4}\left(2\cos\left(\frac{2\pi kp}{n}\right)\left(\sum_{q=0}^{4} 2\cos\left(\frac{2\pi jq}{n}\right) a_q^{(p+1)}\right.\right.$$
$$\left.\left. + \sigma(j)a_{\frac{n}{2}}^{(p+1)} + \phi(j)a_5^{(p+1)}\right)\right) + \left(\sum_{q=0}^{4} 2\cos\left(\frac{2\pi jq}{n}\right) a_q^{(6)} + \sigma(j)a_{\frac{n}{2}}^{(6)} + \phi(j)a_5^{(6)}\right)\phi(k) \tag{8}$$

Notably, the final derivation features the function $\phi(k)$ which is single-variable and therefore easily approximizable despite it's oscillatory behavior and integer domain. Moreover, while the dirichlet kernel is undefined for $k = 0$ and tends towards $n - 9$, any of the given sums with which we've substituted with phi have value $n - 6$ for $k = 0$. $\phi$ is otherwise symmetric and periodic allowing for more analyzable form.

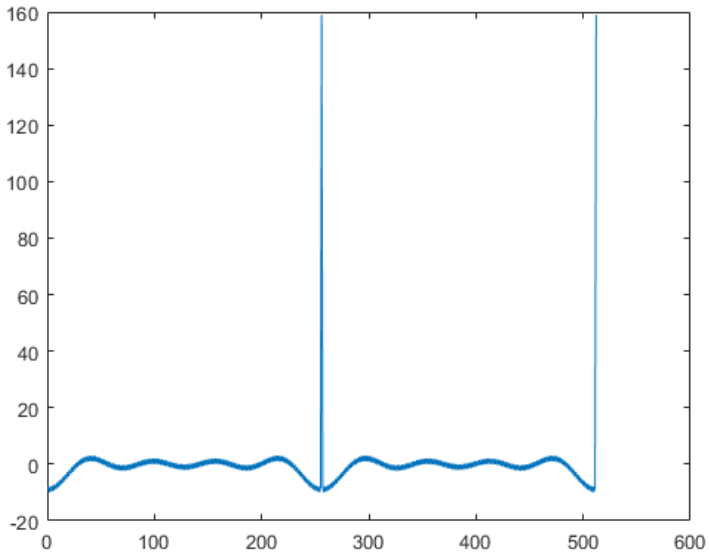



Figure 3: $\phi(x)$ for n=256.

# References


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[4] Alexander Maloney, Daniel A. Roberts, and James Sully. A Solvable Model of Neural Scaling Laws, 2022.

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# Appendix A Appendix

## A.1 Dataset Generation

```
def generate_sample(size, n, m):
    image = np.zeros((size, size), dtype='uint8')
    row = n
    width = 2
    image = cv2.line(image, (0, row), (size - 1, row),
       255, width)

    if row == 0:
        image[size - 1, :] = image[0, :]

    elif row == size - 1:
        image[0, :] = image[size - 1, :]

    col = m
    image = cv2.line(image, (col, 0), (col, size - 1),
       255, width)

```

```
    if col == 0:
        image[:, size - 1] = image[:, 0]
    elif col == size - 1:
        image[:, 0] = image[:, size - 1]

    image = artifical_greyscale(image, n, m, width, size
        )

    return image
```

Listing 1: Dataset Generation

```
def artifical_greyscale(image, row, col, width, size):
    rows = [(row + width) % size, (row - width) % size]
    cols = [(col + width) % size, (col - width) % size]

    grey_value = int(2*255/3)

    for r in rows:
        mask = image[r, :] < 255
        image[r, mask] = grey_value

    for c in cols:
        mask = image[:, c] < 255
        image[mask, c] = grey_value

    return image
```

Listing 2: Artificial Greyscale